Does the Relation Earn Its Label?

Week 11 Tuesday — Relations, equivalence, and partial orders

COMSC 2043 ▪ Discrete Structures and Critical Thinking

October 27, 2026

75 minutes. Spine: Sources, Rules/Assumptions, Work, Check, One-Sentence Summary. Theme for the day: a name is a claim, and a claim needs a witness. Never accept “it feels symmetric.” Ask for the pair.

Today's destination

- Say what a relation is: a set of ordered pairs.
- Test four properties one at a time, with a witness or a counterexample.
- Tell an **equivalence relation** from a **partial order**.
- Catch a confident AI claiming a property it does not have.

Run of show: 0:00 AI Fluency (8). 0:08 definitions (14). 0:22 divisibility model (18). 0:40 student activity (14). 0:54 compare criteria, checkpoint (13). 1:07 exit ticket and preview (18, includes slack).

- "is friends with"
- "has the same birthday as"
- "is a sibling of"
- "beats" in rock–paper–scissors

Gut check, hands up: for each one, is it a fair kind of "sameness"? Then name *one pair* that breaks it.

Do this before any definitions. Expected: (1) friends: friend of a friend need not be a friend, so transitivity fails. (2) same birthday: works, it sorts people into 366 bins. (3) sibling: you are not your own sibling, so reflexivity fails (and half-siblings depending on definition). (4) beats: rock beats scissors, scissors beats paper, paper beats rock. A cycle. Not transitive, not reflexive, not symmetric. Let students argue; do not resolve it, the next slides give the vocabulary.



Eight minutes total for warm-up plus this. Target: symmetric means for every (a,b) in R, (b,a) is in R. The AI showed one pair going forward and never checked the reverse. Test: does 6 divide 2? No. So (2,6) is in R and (6,2) is not. One counterexample kills the claim. Evidence for "for all" needs a proof; evidence against needs one witness.

A **relation** R on a set S is a set of ordered pairs $R \subseteq S \times S$.

- $a \mathrel{R} b$ means $(a, b) \in R$. Order matters: $(2, 6) \neq (6, 2)$.
- Three ways to write the same thing: *list the pairs*, *a matrix*, *a digraph*.
- Name the domain S first. If the domain drifts, the answer changes.

Counting is humbling: on just 3 elements there are $2^9 = 512$ different relations.

Why 512: S x S has 9 ordered pairs and each is in or out. On 2 elements: 16. On 4 elements: 65,536. Most of them are junk; the interesting ones have structure, and the four properties are how we describe structure. Quick poll: how many relations on a 4-element set? (2^{16}.)

Reflexive Does every a relate to itself? $(a, a) \in R$ for all a

Symmetric Does every forward pair have a reverse? $(a, b) \Rightarrow (b, a)$

Antisymmetric Do two-way pairs only happen for $a = b$? $(a, b) \wedge (b, a) \Rightarrow a = b$

Transitive Do chains shortcut? $(a, b) \wedge (b, c) \Rightarrow (a, c)$

- To say **yes**: check *every* pair. To say **no**: show *one* witness.
- Symmetric and antisymmetric are not opposites. Both can hold; neither can hold.

Both hold: equality, R = (1,1),(2,2). Neither: R = (1,2),(2,1),(1,3) on 1,2,3: not symmetric because (1,3) lacks (3,1); not antisymmetric because of (1,2),(2,1). Ask students to build their own “both” and “neither” examples if time.

$a \mathrel{R} b$ when a divides b.

Pairs (9 of them):
$(1,1), (1,2), (1,3), (1,6),$
$(2,2), (2,6), (3,3), (3,6), (6,6)$

Matrix (row a, column b):

	1	2	3	6
1	1	1	1	1
2	0	1	0	1
3	0	0	1	1
6	0	0	0	1



Build the pair list live, row by row from the matrix: ask “which b does 1 divide?” etc. Digraph: arrows are the pairs. Dashed arrow 1 to 6 is the transitive shortcut; loops are reflexivity. Trace the checks on the next slide off this same picture so the matrix, digraph, and list visibly agree.

Property	Holds?	Witness / reason
Reflexive	yes	diagonal of matrix all 1s; $a \mid a$
Symmetric	**no**	$(2,6) \in R$ but $(6,2) \notin R$
Antisymmetric	yes	no off-diagonal pair is two-way (matrix has no mirrored 1s)
Transitive	yes	$(1,2), (2,6) \Rightarrow (1,6)$; all other chains checked

Verdict: reflexive + antisymmetric + transitive $\Rightarrow$ a **partial order**. Not symmetric, so *not* an equivalence relation.

Say the matrix check out loud: symmetric means the matrix equals its mirror across the diagonal. Here 2,6 is a 1 and 6,2 is a 0. Antisymmetric means no off-diagonal 1 is mirrored by another 1. Do not say “divisibility is a partial order” in general without the domain: on all integers including negatives and zero it is trickier; on positive integers it is a partial order. State the domain every time.

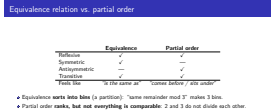
Your turn (14 min): Does the relation earn its label?

Use `student/tuesday-activity.ad`.

1. Represent R as pairs **and** as a matrix or digraph.
2. Classify all four properties. Every "no" needs a counterexample pair or chain.
3. Equivalence, partial order, neither, or both? Name the required properties.
4. Check: does your matrix/digraph agree with your symmetry and antisymmetry answers?

Watch for: domain drift (using $\{1, 2, 3, 4, 5, 6\}$ by accident) and reversed pairs ($b \mid a$ instead of $a \mid b$).

Circulate. Common errors: writing (6,2) because "2 goes into 6" backwards; calling it transitive without checking a chain; saying "both" because it is reflexive. If a pair finishes early: ask them to change the relation to "a and b have the same remainder mod 2" on 1,2,3,6 and re-classify. That one is an equivalence relation.



This is the side-by-side slide the run of show asks for. The "interesting" fact: every equivalence relation is exactly a partition of the set into bins and vice versa. Count check, verified by brute force: on a 3-element set, of the 512 relations, exactly 5 are equivalence relations (the 5 ways to partition 3 things) and exactly 19 are partial orders. On 4 elements: 65,536 relations, 15 equivalences, 219 partial orders.



Answer: a cycle. If A requires B requires C requires A, no valid order exists, which is the same cycle as rock-paper-scissors. Partial orders forbid cycles through antisymmetry plus transitivity. This is the payoff for the warm-up; tie it back explicitly.



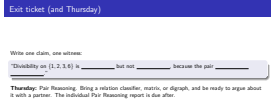
Answers: 4 bins of 3 each. Not a partial order: antisymmetry fails, since 0 4 and 4 0 with 0 not equal to 4. Transitive because equal remainders are an equality, and equality chains. Use as a floater if the activity goes fast, or as a bridge into the checkpoint.

Reasoning Odyssey Checkpoint 2 (Weeks 8–11)

- It **replaces** this week's Decision Gate. Submit one, not both.
- Pull from recursion, logic, probability, and relations: what changed in how you check a claim?
- Use the spine: **Sources** → **Rules** → **Work** → **Check** → **One-Sentence Summary**.

Plan it now (2 min): which one earlier claim of yours would you re-examine with a witness or counterexample?

Details live in student/odyssey-checkpoint-2.md. Have students write a one-line checkpoint plan before they leave: which week, which claim, which test.



Model answer: "reflexive but not symmetric, because the pair (2,6) has no (6,2)." Collect traces. Thursday run of show is in week-11/instructor/thursday-run-of-show.md.